Adaptive Contrastive Learning in Sequential Recommendation based on Perturbation and Restoration Networks

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Abstract

Sequential recommender systems play a vital role in alleviating the challenge of information overload. Although contrastive learning has been increasingly adopted in sequential recommendation to enhance model performance, most existing approaches rely on predefined data augmentation strategies—such as random noise injection or neuron dropout—to generate contrasting views. These strategies, however, often overlook the inherent semantic similarity between the original sequence and its augmented views, which can inadvertently distort user intent and compromise recommendation accuracy. To address this issue, we propose an Adaptive Contrastive Learning framework for Sequential Recommendation (ACLSRec), which incorporates learnable perturbation and restoration networks for adaptive augmentation. The framework dynamically perturbs and restores user representations, thereby ensuring semantic consistency across augmented views and effectively capturing evolving user interest patterns through contrastive learning. Extensive experiments on real-world datasets demonstrate that ACLSRec achieves superior recommendation accuracy compared to several competitive baselines. This work not only establishes a new baseline for sequential recommendation but also paves the way for developing more robust and adaptive contrastive learning frameworks in recommender systems.

CCS Concepts

• **Information systems → Recommender systems**.

Keywords

Sequential Recommendation, Contrastive Learning, Adaptive augmentation, Perturbation and Restoration Networks

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1 Introduction

In today's landscape of information overload, recommender systems have become indispensable tools for helping users discover personalized content efficiently [7, 17]. Among these, sequential recommender systems distinguish themselves by their capacity to model temporal dependencies in user interaction sequences, thereby accurately capturing users' evolving preferences [3, 25, 41]. However, modeling user interaction sequences is primarily challenged by two issues. The first is the inherent sparsity of user behavior sequences. Under cold-start conditions, where users have few interactions, the scarcity of interaction data makes it difficult to model complex interest transitions effectively, thus undermining recommendation precision [15]. The second issue is the presence of implicit noise [34, 35], which can divert a model from capturing genuine temporal patterns, leading to biased or inaccurate recommendations.

Contrastive learning has emerged as a promising approach to mitigate data sparsity and noise interference by leveraging self-supervised signals [26, 39]. The core idea of contrastive learning is to learn discriminative representations by constructing positive and negative sample pairs, with the objective of maximizing the similarity between positive pairs while minimizing that of negative pairs in the feature space. Inspired by this capability, several studies have introduced contrastive learning into sequential recommendation to alleviate data sparsity and noise-related issues [10, 28, 33]. In this context, for a given user, positive samples are typically generated by applying different data augmentation techniques to the user's interaction sequence, while the interaction sequences of other users are treated as negative samples. Ideally, negative samples should be clearly distinguishable from positive samples in the feature space, allowing the model to effectively discriminate between them and enhance sequence representation learning.

However, many contrastive learning-based sequential recommendation methods rely on predefined augmentation strategies, such as random perturbations or dropout, to construct contrastive views [19, 32]. These methods often overlook the semantic consistency between the original sequence and its augmented views,

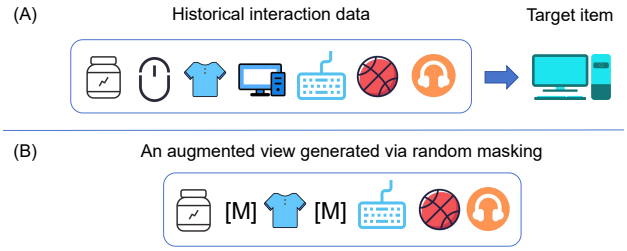


Figure 1: An example of augmentation through random perturbation.

which can compromise the discriminative power of learned representations [12] and ultimately misrepresent genuine user interests. Figure 1 illustrates the problem caused by random perturbation. Subfigure (A) of Figure 1 depicts the original user interaction sequence, which comprises items from two core intents: sports and electronics, with the target item being a desktop computer. Subfigure (B) of Figure 1 illustrates an augmented view generated via random masking. Critically, the masking of key electronic items (e.g., "mouse" and "desktop computer") results in a skewed representation that overemphasizes sports, ultimately leading to a deviation from the user's true intent. Employing dropout to construct augmented views also presents significant challenges. By randomly discarding neurons, this mechanism risks severing essential connections that encode key behavioral features, which can disrupt the learning of temporal dependencies. Consequently, the model may fail to capture rare yet significant patterns, corrupt the semantic integrity of the original sequence, and ultimately misrepresent user interests.

To address the aforementioned limitations, we proposes an Adaptive Contrastive Learning framework for Sequential Recommendation (ACLSRec) in this paper. The framework incorporates learnable perturbation and restoration networks to dynamically perturb and restore user embeddings, thereby ensuring semantic consistency across augmented views while effectively capturing evolving user interest patterns through contrastive learning. By introducing a closed-loop "original → perturbation → restoration" learning constraint, ACLSRec overcomes the semantic bias inherent in conventional augmentation strategies. This design preserves the intrinsic sequential logic of user behaviors while enhancing robustness to sparsity and noise through self-supervised learning. Extensive experiments on real-world datasets demonstrate that ACLSRec achieves superior recommendation accuracy compared to several competitive baselines.

The main contributions of this paper include:

- We construct a dual-path augmentation architecture based on perturbation and restoration networks. This framework enables controllable augmentation and semantic-aware reconstruction of user sequences in the latent space.
- We propose a semantically consistent adaptive contrastive learning paradigm. This approach effectively mitigates the semantic bias introduced by conventional augmentation methods, thereby strengthening representation learning and improving recommendation accuracy.
- Extensive experiments on multiple public datasets demonstrate that ACLSRec significantly outperforms existing contrastive learning baselines in overall recommendation accuracy, validating the effectiveness and generalizability of the proposed framework.

2 Related Work

2.1 Sequential Recommendation

The evolution of sequential recommendation research has undergone a significant paradigm shift, moving from early Markov chain frameworks [20] to deep learning-driven approaches. GRU4Rec [9] innovatively employs Gated Recurrent Units (GRUs) to dynamically model user interaction sequences, providing a novel solution for session-level recommendations. Caser [23] pioneered the use of convolutional neural networks, encoding sequences into matrices to extract local features. The rise of graph neural networks further broadened research perspectives, with models like SR-GNN [31] learning deep item transition relationships by constructing session graphs. The introduction of the Transformer [24] architecture greatly propelled this field, leading to models such as SASRec [11], which applied the self-attention mechanism to sequential recommendation. BERT4Rec [21] adopted the pre-training paradigm from natural language processing, using bidirectional self-attention and masked prediction tasks to capture comprehensive contextual information. Additionally, SSE-PT [30] integrated user embeddings into the Transformer framework, offering a new approach to personalized sequential recommendations.

2.2 Contrastive Learning for Sequential Recommendation

As an efficient self-supervised learning paradigm [8], contrastive learning has shown significant advantages in fields such as computer vision [13] and natural language processing [4], and is extensively employed to mitigate data sparsity and noise issues in recommender systems [22, 33]. CL4SRec [32] constructs augmented views of user interaction sequences through random cropping, masking, and rearrangement, and simultaneously optimizes the contrastive learning objective with the recommendation task. S3-rec [40] enhances representation quality through the integration of mutual information maximization and pre-training. DuoRec [19] proposes a model-level augmentation strategy, generating contrastive samples via Dropout perturbation to mitigate representation degradation. TGCL4SR [36] constructs a temporal graph contrastive learning framework to collaboratively learn global and local features of sequences.

Predefined augmentation strategies often overlook the semantic consistency between the original sequence and its augmented views, which can compromise the discriminative power of learned representations [12]. Some semantics-preserving augmentation methods are proposed in sequential recommendation. MCLRec [18] achieves adaptive augmentation by combining the meta-optimization mechanism with the learnable augmentation module. ICLRec [1] incorporates user intent modeling, and designs intent-aware comparison tasks to enhance the consistency of representation between views with the same semantic intention. ELCRec[14] integrates representation learning into an end-to-end learnable clustering framework

and proposes intent-assisted contrastive learning by using cluster centers as self-supervision signals. LMA4Rec [6] adaptively generates contrastive views using learnable Bernoulli Dropout, ensuring the preservation of sequence semantics. RCL [27] uses similar sequences as additional positive samples and introduce a Relative Contrastive Learning (RCL) framework for sequential recommendation. CFIT4Rec [37] constructs augmented samples from the frequency domain, which allows the sequence encoder to accommodate different frequency components and improve its inference ability.

3 Notation and Problem Description

We denote the set of users as U and the set of items as V, where $|U|$ and $|V|$ represent the number of users and items, respectively. Each user is denoted by $u \in U$, and each item by $v \in V$. In sequential recommendation, a user's historical behaviors are typically ordered chronologically. Thus, the interaction sequence of user u is represented as $s_u = [v_1^u, v_2^u, \ldots, v_{|s_u|}^u]$, where v_t^u denotes the item that u interacted with at time step t, and $|s_u|$ is the length of the sequence. The subsequence $s_{u,t} = [v_1^u, v_2^u, \ldots, v_t^u]$ refers to the set of items that user u interacted with before time step $t+1$.

This paper focuses on the key challenge of next-item prediction using only user interaction sequences. The primary task is to predict the next item that user u is most likely to interact with at time step $|s_u|+1$, based on the interaction sequence $s_u = [v_1^u, v_2^u, \ldots, v_{|s_u|}^u]$. The problem can be formally defined as follows:

$$v_u^* = \arg\max_{v_i \in V} P(v_{|s_u|+1}^u = v_i | s_u), \tag{1}$$

where $P(v_{|s_u|+1}^u = v_i | s_u)$ corresponds to the conditional probability of user u interacting with item v_i at the next time step $|s_u|+1$, given the historical sequence s_u. v_u^* denotes the final recommended item, which is the one assigned the highest probability by the recommendation model.

4 Methodology

4.1 Overall Framework

The proposed Adaptive Contrastive Learning framework for Sequential Recommendation (ACLSRec) incorporates learnable perturbation and restoration networks to dynamically perturb and restore user embeddings, thereby ensuring semantic consistency across augmented views while effectively capturing evolving user interest patterns through contrastive learning. The overall framework is shown in Figure 2. It includes three modules: user sequence encoder, perturbation and restoration networks, and multi-task learning. To ensure uniform processing, each user's historical interaction sequence is standardized to a canonical length T. For a user u at time step $t+1$, if the sequence length $t > T$, only the most recent T interactions are retained, i.e., $s_{u,t} = [v_{t-T+1}^u, v_{t-T+2}^u, \ldots, v_t^u]$. If $t < T$, the sequence is left-padded with placeholder items to achieve length T [11, 31]. For notational simplicity, we uniformly represent the processed interaction sequence of user u as $s_u = [v_1^u, v_2^u, \ldots, v_T^u]$.

4.2 User Sequence Encoder

To effectively model evolving user preferences and the complex transition patterns in interaction sequences, we employ a Transformer-based architecture as core sequence encoder. The self-attention mechanism of Transformer is particularly advantageous for adaptively weighting relevant behaviors and capturing long-range dependencies within a sequence. The user sequence encoder, which operates on the fixed-length sequence, comprises two primary components: an embedding layer that maps items into a dense vector space, and a multi-layer Transformer that contextualizes these items.

4.2.1 Embedding Layer. In the embedding layer, given the user interaction sequence s_u, we initially map each item to a latent space:

$$\boldsymbol{E}_t = \boldsymbol{W}_e[v_t^u], \quad \boldsymbol{W}_e \in \mathbb{R}^{|V| \times d}, \tag{2}$$

where $\boldsymbol{W}_e$ is the item embedding matrix, t is the time step index, and d is the dimension of the latent vector space.

To preserve the positions information of items in a sequence, we incorporate a learnable positional embedding into the model:

$$\boldsymbol{P}_t = \boldsymbol{W}_p[t], \quad \boldsymbol{W}_p \in \mathbb{R}^{T \times d}, \tag{3}$$

where $\boldsymbol{W}_p$ is the positional embedding matrix.

The positional embedding and item embedding are fused to form the input embedding for every item in a sequence, and the input embeddings of all items in a sequence are stacked to form the input embedding matrix of the sequence:

$$\boldsymbol{H} = \begin{bmatrix} \boldsymbol{E}_1 + \boldsymbol{P}_1 \\ \boldsymbol{E}_2 + \boldsymbol{P}_2 \\ \vdots \\ \boldsymbol{E}_T + \boldsymbol{P}_T \end{bmatrix}, \quad \boldsymbol{H} \in \mathbb{R}^{T \times d}. \tag{4}$$

The combined input embedding matrix $\boldsymbol{H}$ first undergoes layer normalization to stabilize training, followed by dropout for regularization:

$$\boldsymbol{H}_{in} = Dropout(LayerNorm(\boldsymbol{H})). \tag{5}$$

This yields the final input representation $\boldsymbol{H}_{in}$, which is then fed into the subsequent transformer layers.

4.2.2 Multi-layer Transformer. The core of user sequence encoder is a multi-layer Transformer with L identical layers. Each Transformer layer takes the output from the previous layer as its input. To extract the information from different subspaces at each position, we employ multi-head self-attention instead of a single attention function. Multi-head attention uses different linear projections to project the input representations into k subspaces. It then applies the self-attention mechanism to each head, concatenates the outputs of all heads, and fuses them through a linear layer:

$$\begin{aligned} MH(\boldsymbol{H}^l) &= concat(\boldsymbol{head}_1; \boldsymbol{head}_2; ...; \boldsymbol{head}_k)\boldsymbol{W}^o \\ \boldsymbol{head}_i &= Attention(\boldsymbol{H}^l\boldsymbol{W}_i^Q, \boldsymbol{H}^l\boldsymbol{W}_i^K, \boldsymbol{H}^l\boldsymbol{W}_i^V), \end{aligned} \tag{6}$$

where $\boldsymbol{W}_i^Q, \boldsymbol{W}_i^K, \boldsymbol{W}_i^V \in \mathbb{R}^{d \times \frac{d}{k}}$, and $\boldsymbol{W}^o \in \mathbb{R}^{d \times d}$ are the weight matrices. $\boldsymbol{H}^l$ is the output of the l-th layer. The attention mechanism is implemented through the following dot product operation:

$$Attention(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = soft\max(\frac{\boldsymbol{Q}\boldsymbol{K}^\top}{\sqrt{d/k}})\boldsymbol{V}. \tag{7}$$

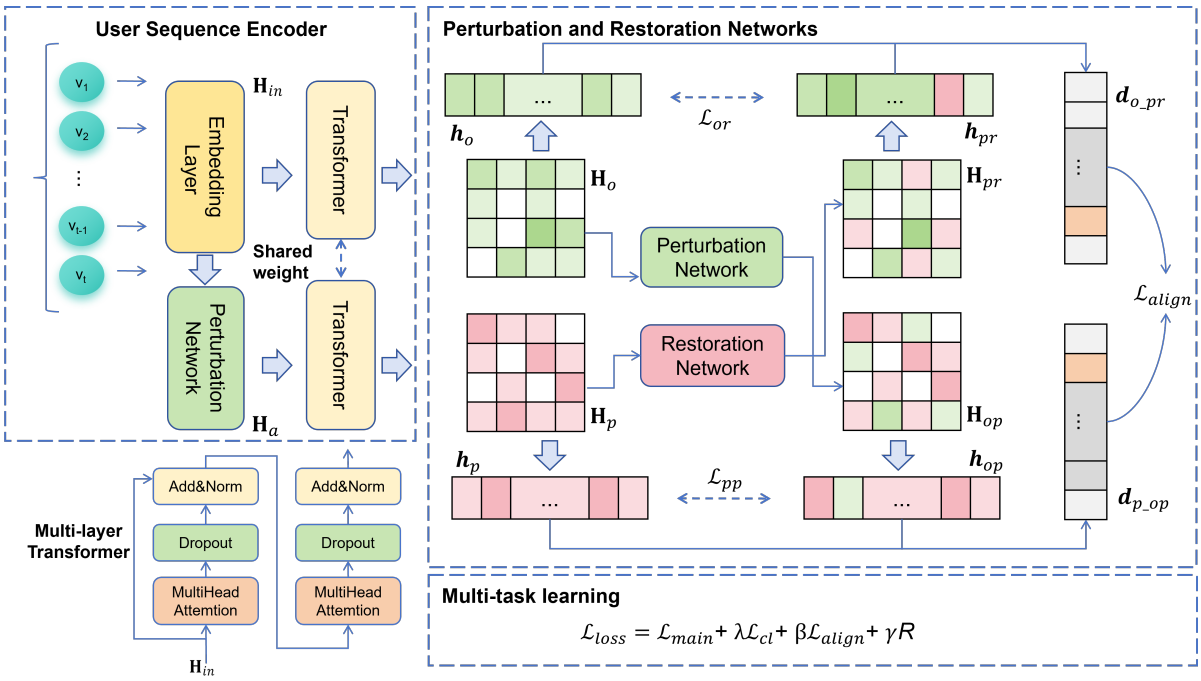


Figure 2: The overall framework of ACLSRec. It contains four modules: three modules: user sequence encoder, perturbation and restoration networks, and multi-task learning.

In sequential recommendation, only the information prior to the current time step t can be used when predicting the next item. Therefore, we apply a masking operation to the attention mechanism. If $j > i$, the relationship between $\boldsymbol{Q}_i$ and $\boldsymbol{K}_j$ is ignored. This masking ensures that each position i attends only to position i and earlier positions in the sequence, thereby preserving the autoregressive property of the model.

The output $MH(\boldsymbol{H}^l)$ of the multi-head self-attention layer is then fed into a position-wise feed-forward network, which applies a nonlinear transformation and further refines the feature representations:

$$FFN(\boldsymbol{H}^l) = GELU(MH(\boldsymbol{H}^l)\boldsymbol{W}_1 + \boldsymbol{b}_1)\boldsymbol{W}_2 + \boldsymbol{b}_2, \tag{8}$$

where $\boldsymbol{W}_1$ and $\boldsymbol{W}_2$ are weight matrices, and $\boldsymbol{b}_1$ and $\boldsymbol{b}_2$ are bias vectors. This feed-forward network enhances the model's ability to capture complex patterns and interactions within the sequence.

To stabilize and accelerate training, a Dropout layer is applied sequentially after each feed-forward Network, followed by a residual connection and layer normalization (Add & Norm):

$$\boldsymbol{H}^{l+1} = LayerNorm(\boldsymbol{H}^l + Dropout(FFN(\boldsymbol{H}^l))). \tag{9}$$

The output matrix from the final layer is denoted as $\boldsymbol{H}^L$. As the objective of sequential recommendation is to predict the next item based on a user's interaction history, a user's final representation is defined as her preference representation at time step T, which corresponds to the last row of the output matrix $\boldsymbol{H}^L$:

$$\boldsymbol{h} = \boldsymbol{H}^L[-1] \in \mathbb{R}^d. \tag{10}$$

The multi-layer Transformer allows the model to capture complex, long-range dependencies within the user sequence, understanding the context and relationships between items.

4.3 Perturbation and Restoration Networks

To mitigate the semantic bias introduced by conventional augmentation techniques in contrastive learning, we design a dual-path augmentation architecture and propose a semantically consistent adaptive contrastive learning paradigm based on perturbation and restoration networks. The learnable perturbation network intentionally introduces controlled noise or perturbations to generate augmented views in the latent space. In contrast, the restoration network is employed to reconstruct these representations, ensuring semantic consistency.

4.3.1 Dual-path Augmentation Architecture. The dual-path architecture processes the input through two parallel streams. In one branch, the input sequence embedding matrix $\boldsymbol{H}_{in}$ (from Equation 5) is fed into the perturbation network P. This network, a Multilayer Perceptron (MLP) parameterized by θ_p, transforms the input to produce the perturbation-augmented sequence embedding matrix $\boldsymbol{H}_a$:

$$\boldsymbol{H}_a = \text{MLP}_{\theta_p}(\boldsymbol{H}_{in}). \tag{11}$$

The matrix $\boldsymbol{H}_a$ is then fed into the multi-layer Transformer for deep feature encoding, producing the output matrix $\boldsymbol{H}_p$:

$$\boldsymbol{H}_p = \text{Transformer}(\boldsymbol{H}_a). \tag{12}$$

The matrix $\boldsymbol{H}_p$ represents the features learned from the augmented view generated by the perturbation network. To preserve semantic

consistency, the restoration network is introduced to restore the features representation. The restoration network is a Multilayer Perceptron (MLP) parameterized by θ_r. The restored representation matrix is formulated as follows:

$$\boldsymbol{H}_{pr} = \mathrm{MLP}_{\theta_r}(\boldsymbol{H}_p), \tag{13}$$

In the other branch, the input sequence embedding matrix $\boldsymbol{H}_{in}$ (from Equation 5) is fed directly into the multi-layer Transformer for deep feature encoding, producing the output matrix $\boldsymbol{H}_o$:

$$\boldsymbol{H}_o = \mathrm{Transformer}(\boldsymbol{H}_{in}). \tag{14}$$

To make the perturbation-augmented view generated by the perturbation network P controllable, we introduce another perturbation network P' to generate contrastive views:

$$\boldsymbol{H}_{op} = \mathrm{MLP}_{\theta_{p'}}(\boldsymbol{H}_o). \tag{15}$$

4.3.2 Contrastive Learning. The perturbation network adds controlled variations to user sequences to simulate real-world noise, while the restoration network ensures the model can faithfully reconstruct the original semantics. These networks are trained using contrastive learning to minimize the distance between perturbation-augmented views while preserving the accuracy of the restored representation. This approach enables the encoder to learn robust, aligned, and high-quality user representations for recommendation.

The last rows of each representation matrix are considered as the user representations:

$$\boldsymbol{h}_* = \boldsymbol{H}_*[-1], \quad * \in \{p, pr, o, op\}. \tag{16}$$

The model's contrast loss comprises two components: the InfoNCE loss [5] between the original representation $\boldsymbol{h}_o$ and the restored representation $\boldsymbol{h}_{pr}$, and the InfoNCE loss between the two perturbation representations $\boldsymbol{h}_p$ and $\boldsymbol{h}_{op}$. These losses operate independently and are formulated as:

$$\mathcal{L}_{or} = -\frac{1}{|U|}\sum_{u\in U}\log\frac{\exp\left(\boldsymbol{h}_o^u(\boldsymbol{h}_{pr}^u)^T/\tau\right)}{\sum_{v\in U_u}\exp\left(\boldsymbol{h}_o^u(\boldsymbol{h}_{pr}^v)^T/\tau\right)+\sum_{v\in U_u}\exp\left(\boldsymbol{h}_o^v(\boldsymbol{h}_{pr}^u)^T/\tau\right)} \tag{17}$$

and

$$\mathcal{L}_{pp} = -\frac{1}{|U|}\sum_{u\in U}\log\frac{\exp\left(\boldsymbol{h}_p^u(\boldsymbol{h}_{op}^u)^T/\tau\right)}{\sum_{v\in U_u}\exp\left(\boldsymbol{h}_p^u(\boldsymbol{h}_{op}^v)^T/\tau\right)+\sum_{v\in U_u}\exp\left(\boldsymbol{h}_p^v(\boldsymbol{h}_{op}^u)^T/\tau\right)}, \tag{18}$$

where τ is the temperature parameter. $\mathcal{L}_{or}$ aligns $\boldsymbol{h}_o^u$ with $\boldsymbol{h}_{pr}^u$ for each user u using in-batch negatives across other users' pairs, while $\mathcal{L}_{pp}$ aligns $\boldsymbol{h}_p^u$ with $\boldsymbol{h}_{op}^u$ for each user u analogously. U_u denotes the set of users in the same batch as user u. Although the two loss functions seem similar, they serve distinct purposes. The loss function $\mathcal{L}_{or}$ promotes consistency across augmented views, thereby reinforcing the model's stability and generalization. Conversely, the loss function $\mathcal{L}_{pp}$ enhances the model's robustness by maintaining performance in the presence of noise or augmentation. The contrastive loss is calculated by the sum of $\mathcal{L}_{or}$ and $\mathcal{L}_{pp}$:

$$\mathcal{L}_{cl} = \mathcal{L}_{or} + \mathcal{L}_{pp}. \tag{19}$$

4.3.3 Distance Constraint. To achieve semantic controllability in the augmentation process and mitigate the impact of dimensional differences on distance calculations, we propose a distance constraint loss based on cross-path symmetry constraints.

We apply L2 normalization to each user representation ($\boldsymbol{h}_p$, $\boldsymbol{h}_{pr}$, $\boldsymbol{h}_o$, $\boldsymbol{h}_{op}$) using:

$$\tilde{\boldsymbol{h}}_* = \frac{\boldsymbol{h}_*}{\|\boldsymbol{h}_*\|_2}, \quad * \in \{p, pr, o, op\}. \tag{20}$$

Two cross-view geometric distance metrics are defined by the normalized representation:

$$\boldsymbol{d}_{o_pr} = (1 - \tilde{\boldsymbol{h}}_o^T\tilde{\boldsymbol{h}}_{pr})/\tau \tag{21}$$

and

$$\boldsymbol{d}_{p_op} = (1 - \tilde{\boldsymbol{h}}_p^T\tilde{\boldsymbol{h}}_{op})/\tau, \tag{22}$$

where τ is the temperature coefficient that scales the geometric distance to match the magnitude of the contrastive loss. By calculating the average distance, a symmetric distance constraint loss function is constructed:

$$\mathcal{L}_{align} = E\left[(\boldsymbol{d}_{o_pr} - \boldsymbol{d}_{p_op})^2\right], \tag{23}$$

where E means calculating the average value.

4.3.4 Regularization Constraint. To prevent the perturbation and restoration networks from generating semantically collapsed representations and to avoid over-alignment in the representation space, we introduces a dynamic contrast regularization mechanism. This approach adaptively adjusts the decision boundary for positive and negative sample pairs by analyzing their similarity distribution within a batch.

For each user u in a batch, the user representations $\boldsymbol{h}_o^u \in \mathbb{R}^d$ and $\boldsymbol{h}_{pr}^u \in \mathbb{R}^d$ are stacked vertically to form an extended matrix $\boldsymbol{Z}$, as defined in Equation 24. The same concatenation operation is applied to the item representations $\boldsymbol{h}_p$ and $\boldsymbol{h}_{op}$ respectively.

$$\boldsymbol{Z} = \begin{bmatrix} \boldsymbol{h}_o^1 \\ \boldsymbol{h}_o^2 \\ \vdots \\ \boldsymbol{h}_o^N \\ \boldsymbol{h}_{pr}^1 \\ \boldsymbol{h}_{pr}^2 \\ \vdots \\ \boldsymbol{h}_{pr}^N \end{bmatrix} \in \mathbb{R}^{2N\times d}, \tag{24}$$

where N is the batch size. The full similarity matrix is calculated as:

$$\boldsymbol{S} = \boldsymbol{Z}\boldsymbol{Z}^T \in \mathbb{R}^{2N\times 2N}. \tag{25}$$

Then, the binary mask matrix $\boldsymbol{M} \in \{0,1\}^{2N\times 2N}$ is defined as:

$$\boldsymbol{M}_{ij} = \begin{cases} 1, & \text{if } i = j \cup |i-j| = N \\ 0, & \text{other} \end{cases}. \tag{26}$$

This matrix is used to distinguish positive pairs from negative pairs across users, serving as a foundation for calculating dynamic boundaries. The similarity value sets for positive pairs and negative pairs can be represented as:

$$\sigma^+ = \{\boldsymbol{S}_{i,i+N} \mid i = 1, \ldots, N\}, \tag{27}$$

and

$$\sigma^- = \{\boldsymbol{S}_{i,j} \mid \boldsymbol{M}_{i,j} = 0\}. \tag{28}$$

The decision boundary is dynamically adjusted based on the current batch data distribution to prevent underfitting or overfitting

that may result from a fixed threshold. The dynamic boundary is established as follows:

$$o_{min} = \min(\min(\sigma^+), \max(\sigma^-)), \tag{29}$$

$$o_{max} = \max(\min(\sigma^+), \max(\sigma^-)). \tag{30}$$

Then, we use the dynamic boundary to define the regularization term:

$$R = \frac{1}{|\sigma^+|} \sum_{s \in \sigma^+} \max(s - o_{min}, 0) + \frac{1}{|\sigma^-|} \sum_{s \in \sigma^-} \max(o_{max} - s, 0). \tag{31}$$

4.4 Multi-task learning

To leverage contrastive learning to improve the performance of sequential recommendation, we adopt a multi-task strategy to jointly optimize the main sequence prediction task and the additional contrastive learning task.

We use the negative log-likelihood calculated by binary cross entropy as the main loss function for each user u at each time step t:

$$\mathcal{L}_{main}(s_{u,t}) = -\log \frac{\exp(\boldsymbol{h}_o^u(\boldsymbol{W}_e[v_{t+1}^+])^\top)}{\exp(\boldsymbol{h}_o^u(\boldsymbol{W}_e[v_{t+1}^+])^\top) + \sum_{v_{t+1}^- \in V^-} \exp(\boldsymbol{h}_o^u(\boldsymbol{W}_e[v_{t+1}^-])^\top)}, \tag{32}$$

where $\boldsymbol{W}_e$ is the item embedding matrix, $\boldsymbol{h}_o^u$ is the learned representation of user u at time step t, v_{t+1}^+ represents the item that user u actually interacts with at time step $t+1$, and v_{t+1}^- represents the negative items selected from the negative item set V^-. This loss directly optimizes the model's accuracy in predicting the immediate next item in a user's interaction sequence.

The total loss is a weighted sum of the main task loss, the contrastive loss, the distance constraint loss, and the regularization term:

$$\mathcal{L}_{loss} = \mathcal{L}_{main} + \lambda \mathcal{L}_{cl} + \beta \mathcal{L}_{align} + \gamma R, \tag{33}$$

where λ, β, and γ are hyperparameters balancing the contribution of different loss components.

4.5 Time Complexity Analysis

We analyze the computational complexity of ACLSRec from theoretical perspectives. The overall time complexity mainly includes the following parts:

(1) The Transformer encoder. The complexity of base Transformer encoder is $O(T^2d + Td^2)$, where T is sequence length and d is hidden dimension.

(2) Contrastive learning module. The framework uses dual-path augmentation architecture to generate augmented views for each user. In each branch, we obtain $2B$ representations (two views per sequence), where B is the batch size. The complexity of computing the similarity matrix is $O(2(2B)^2d) = O(B^2d)$.

(3) Distance constraint loss. This loss calculates the alignment between two view representations of the same sequence, with a complexity of O(Bd).

(4) Perturbation and restoration networks. All the networks are based MLP. A 2-layer MLP projection contributes minimal $O((Td)^2)$ additional cost.

Thus, the total training complexity for one batch is: $O(B^2d + B(Td)^2)$.

5 Experiment

5.1 Experimental Setup

5.1.1 Dataset. We conducted experiments on three public datasets sourced from the Amazon e-commerce platform [16], specifically from the Beauty, Sports, and Toys domains. During preprocessing, we binarized all user interactions: the presence of a rating or review was encoded as '1', and its absence as '0'. For each user, we removed duplicate interactions and chronologically sorted the remaining ones to construct their interaction sequences. To ensure data quality, we retained only the "5-core" subset by iteratively filtering out users and items with fewer than five interactions. The statistics of the processed datasets are summarized in Table 1.

Table 1: Dataset Statistics (After Preprocessing)

Dataset	users	items	actions	avg.len	density
Beauty	22363	12101	198502	8.8	0.07%
Sports	35598	18357	296337	8.3	0.05%
Toys	19412	11917	167597	8.6	0.07%

5.1.2 Evaluation Metrics. This paper employs the leave-one-out [29] method to assess the performance of the recommendation model. This evaluation paradigm is widely used in sequential recommendation research. For each user's interaction sequence, the last interaction is designated as the test set, the second-to-last as the validation set, and the remaining interactions form the training set. To ensure an unbiased evaluation, we included every item not in a user's interaction history as a candidate during ranking. This strategy thereby preserves the integrity and comparability of all reported metrics. The experiment utilizes the following three widely recognized evaluation metrics:

- **Hit Rate (HR@N)**: Measures whether the target item is present in the Top-N recommendation list. It reflects the model's ability to include relevant items in the shortlist, with higher values indicating better recall performance.
- **Normalized Discounted Cumulative Gain (NDCG@N)**: Evaluates the overall ranking quality by considering the position of relevant items. It assigns higher weights to items ranked at the top through a logarithmic discount, providing a normalized score that more comprehensively reflects the recommendation list quality. Higher values indicate better ranking accuracy.
- **Mean Reciprocal Rank (MRR@N)**: Focuses on the rank of the target item by calculating the average reciprocal of its position within the Top-N list. This metric emphasizes the importance of placing the user's preferred items at the top, with higher values indicating superior ranking performance.

In this experiment, we set $N \in \{5, 10\}$ to evaluate the model's performance across recommendation lists of different lengths.

Table 2: The recommendation performance of different models. The bold and underlined scores denote the best and sub-optimal results, respectively.

Dataset	Metric	SASRec	BERT4Rec	S3RecMP	CL4SRec	DuoRec	CFIT4SRec	MCLRec	ECLRec	ACLSRec	Improve
Beauty	HR@5	0.0369	0.0364	0.0382	0.0406	0.0547	0.0546	0.0566	0.0505	**0.0582**	2.83%
	HR@10	0.0608	0.0583	0.0634	0.0663	0.0843	0.0786	0.0844	0.0723	**0.0846**	0.24%
	NDCG@5	0.0213	0.0228	0.0244	0.0228	0.0345	0.0339	0.0341	0.0345	**0.0388**	12.46%
	NDCG@10	0.0290	0.0307	0.0335	0.0311	0.0440	0.0416	0.0431	0.0415	**0.0469**	6.59%
	MRR@5	0.0162	0.0175	0.0182	0.0169	0.0278	0.0271	0.0267	0.0292	**0.0324**	10.96%
	MRR@10	0.0193	0.0208	0.0215	0.0203	0.0318	0.0303	0.0304	0.0321	**0.0357**	11.21%
Sports	HR@5	0.0233	0.0217	0.0209	0.0248	0.0300	0.0298	0.0279	0.0282	**0.0339**	13.00%
	HR@10	0.0386	0.0359	0.0322	0.0396	0.0458	0.0459	0.0448	0.0434	**0.0497**	8.28%
	NDCG@5	0.0126	0.0143	0.0137	0.0138	0.0192	0.0172	0.0172	0.0181	**0.0218**	13.54%
	NDCG@10	0.0176	0.0190	0.0173	0.0186	0.0244	0.0223	0.0227	0.0230	**0.0269**	10.25%
	MRR@5	0.0091	0.0102	0.0098	0.0102	0.0122	0.0130	0.0137	0.0148	**0.0179**	20.95%
	MRR@10	0.0111	0.0123	0.0119	0.0122	0.0143	0.0151	0.0160	0.0160	**0.0199**	24.38%
Toys	HR@5	0.0635	0.0371	0.0461	0.0643	0.0722	0.0717	0.0497	0.0589	**0.0732**	1.39%
	HR@10	0.0809	0.0524	0.0680	0.0817	0.0910	0.0900	0.0658	0.0834	**0.0924**	1.54%
	NDCG@5	0.0478	0.0259	0.0314	0.0486	0.0538	0.0540	0.0342	0.0402	**0.0550**	1.85%
	NDCG@10	0.0532	0.0309	0.0385	0.0542	0.0599	0.0599	0.0394	0.0481	**0.0612**	2.17%
	MRR@5	0.0425	0.0275	0.0335	0.0433	0.0477	0.0481	0.0291	0.0341	**0.0490**	1.87%
	MRR@10	0.0446	0.0312	0.0378	0.0456	0.0502	0.0505	0.0313	0.0373	**0.0515**	1.98%

5.1.3 Baseline. To ensure a comprehensive evaluation, We compare ACLSRec with several representative sequential recommendation algorithms, including self-attention based models, contrastive-based models with conventional augmentation, and contrastive-based models with semantics-preserving augmentation.

(1) Self-attention based Models

SASRec [11]: A sequential recommendation model based on the self-attention mechanism, which effectively captures the dynamic interest patterns of users.

BERT4Rec [21]: A sequential recommendation model based on BERT [2], which utilizes bidirectional Transformer encoders and a masked sequence modeling objective to capture complex dependencies in user interaction sequences.

(2) Contrastive-based Models with Conventional Augmentation

S3Rec [40]: A self-supervised sequential recommendation framework that designs multiple pre-training tasks, such as masked item prediction, sequence recovery, and next-item prediction, to enhance sequential representation learning.

CL4SRec [32]: Employs contrastive learning with data augmentations such as item cropping, masking, and reordering to enrich sequential representations.

DuoRec [19]: A contrastive learning framework designed for the representation degradation problem, which utilizes a Dropout-driven model-level augmentation and hard positive sampling.

(3)Contrastive-based Models with Semantics-preserving Augmentation

CFIT4Rec [37]: Applies three augmentation operations in the frequency-domain for contrastive learning, which allows the sequence encoder to accommodate different frequency components and improve its inference ability.

MCLRec [18]: A meta-optimization contrastive learning framework that adopts a dual-view (data and learnable model) augmentation strategy to improve feature discrimination.

ELCRec [14]: A novel intent learning method for sequential recommendation. it integrates representation learning into an end-to-end learnable clustering framework for effective and efficient recommendation.

5.1.4 Implementation Details. The proposed ACLSRec model was developed within the RecBole framework [38]. Model configurations were set as follows: the embedding dimension was 64, and the architecture comprised a 2-layer Transformer with 2 attention heads per layer. We employed the Adam optimizer with a learning rate of 0.001 and a batch size of 256 for training. An early stopping strategy was applied, which halted the training if the MRR@10 metric did not improve for 10 consecutive epochs. For hyperparameter tuning, we conducted a grid search over the following candidate sets: $\lambda \in \{0.01, 0.02, 0.05, 0.1, 0.2\}$, $\beta \in \{0.1, 0.2, 0.5, 1, 2, 5\}$, $\gamma \in \{0.001, 0.002, 0.005, 0.01, 0.02\}$, and $\tau \in \{0.05, 0.1, 0.2, 0.5, 1\}$. All experiments are completed on the NVIDIA GeForce RTX 4060 Ti GPU.

5.2 Overall Performance

Experiments were conducted across all three datasets to compare the recommendation performance of ACLSRec against multiple baselines, and the results were obtained using the optimal parameters reported for each respective model. As presented in Table 2, the proposed ACLSRec method consistently achieves state-of-the-art performance across all three datasets, demonstrating significant advantages over the existing baselines. ACLSRec demonstrates its most substantial gains on the Sports dataset, with remarkable improvements of 24.38% in MRR@10 and 20.95% in MRR@5 over the strongest baselines. Notable improvements are also observed in NDCG@5 (13.54%) and HR@5 (13.00%). On the Beauty dataset, the method shows strong improvements in ranking accuracy, with 12.46% and 6.59% gains in NDCG@5 and NDCG@10, respectively, and approximately 11% gains in both MRR@5 and MRR@10. The

improvements on the Toys dataset are more modest, generally around 1.3-2.2% across all metrics.

The performance improvements vary significantly across different metrics and datasets. The most dramatic improvements are often seen in MRR, especially on the Sports dataset. MRR is highly sensitive to the position of the first relevant item in the recommendation list, and it rewards models that rank the target item highly. The substantial boost in MRR suggests that ACLSRec's contrastive learning mechanism is particularly effective at learning high-quality user representations that push the single most relevant item to the very top of the list. In contrast, HR@K is a recall-based metric that only checks for the presence of a relevant item within the top-K, regardless of its rank. Thus, a minor enhancement in HR (e.g., 0.24% in Beauty HR@10) coupled with a significant boost in MRR suggests that ACLSRec excels in accurately ranking the most relevant items at the top of the recommendation list. This is a critical advantage in real-world systems where user attention is focused on the first few recommendations. ACLSRec performs excellently on the Sports dataset. That may because the interaction in Sports dataset is sparser, and ACLSRec is particular effectiveness in sparse scenarios. In contrast, Toys dataset is dense with a high baseline, resulting in a naturally smaller improvement margin.

5.3 Ablation Study

In order to systematically verify the effectiveness of each core component in ACLSRec, we conducted comprehensive ablation studies on two datasets, Beauty and Sports. As summarized in Table 3, we designed four experimental configurations: (A) the complete ACLSRec model; (B) removing the distance constraint loss (w/o L_{align}); (C) removing the contrastive loss (w/o L_{pp} and L_{rr}); and (D) removing all the key components (equivalent to SASRec baseline).

The experimental results clearly demonstrate that the fully-configured ACLSRec model (A) achieves optimal performance across all evaluation metrics and datasets, validating the completeness of our architectural design. Notably, the comparison between configurations (A) and (C) reveals that the contrastive loss contributes substantially to model performance. This significant enhancement demonstrates the effectiveness of the proposed contrastive learning paradigm based on perturbation and restoration networks. Furthermore, by comparing configurations (A) and (B), we observe that the distance constraint loss provides additional performance gain. This finding confirms that the distance constraint effectively balances representation distributions during contrastive learning.

Table 3: Ablation experiments on different comparison items

Model	Beauty		Sports	
	HR@5	NDCG@5	HR@5	NDCG@5
(A) ACLSRec	**0.0582**	**0.0388**	**0.0339**	**0.0218**
(B) w/o L_{align}	0.0495	0.0323	0.0297	0.0189
(C) w/o L_{pp} and L_{rr}	0.0384	0.0241	0.0229	0.0136
(D) SASRec	0.0369	0.0213	0.0233	0.0126

5.4 Parameter Influence

We systematically evaluated the impact of three loss weight: contrastive loss weight λ, distance constraint loss weight β, and regularization weight γ on three evaluation metrics (HR@5, MRR@5, and NDCG@5). Figure 3 presents a comprehensive sensitivity analysis of these three hyperparameters on model performance across the Beauty and Sports datasets. Analysis of the contrastive loss weight λ shows a consistent trend for both datasets: performance peaks at $\lambda = 0.1$ and falls off with larger values. This confirms that a well-calibrated λ is crucial, as an overly strong contrastive signal drowns out the main learning objective. The distance constraint loss weight β exhibits more complex effects on model performance. Optimal results are observed at $\beta = 1$ for both datasets, with performance staying robust for values exceeding 1, indicating strong parameter resilience. Regularization weight also demonstrates significant impact on model performance. Optimal results are observed at $\gamma = 0.005$ for both datasets. Overly strong regularization leads to underfitting, whereas insufficient regularization may cause overfitting.

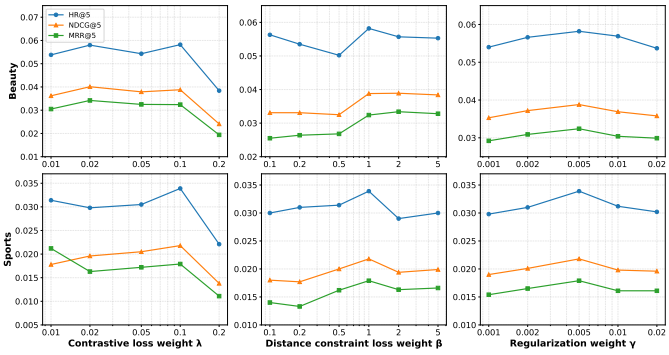


Figure 3: Performance of the ACLSRec model with different loss weight λ, β, γ.

Figure 4 illustrates the impact of the temperature parameter τ on model performance across the Beauty and Sports datasets. Optimal results are observed at $\tau = 0.1$ for both datasets. The results demonstrate a consistent non-monotonic relationship where all three metrics—HR@5, MRR@5, and NDCG@5—initially improve with increasing τ, peak at moderate settings, and then decline with further increases. This pattern confirms that an optimal value effectively sharpens the similarity distribution for contrastive learning, while values that are too low or too high result in suboptimal gradient scaling and diminished recommendation performance.

6 Conclusion

Traditional sequential recommendation models often struggle with noisy and sparse interaction data, while existing contrastive learning methods tend to introduce semantic bias during augmentation, limiting their ability to generalize. In this paper, we proposed Adaptive Contrastive Learning framework for Sequential Recommendation (ACLSRec), a novel sequential recommendation model designed to more effectively capture users' dynamic and evolving

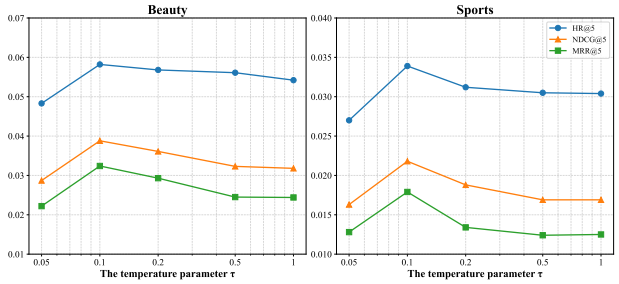


Figure 4: Performance of the ACLSRec model with different temperature parameter τ.

preferences. By integrating a dual-path augmentation architecture and a semantically consistent adaptive contrastive learning paradigm based on perturbation and restoration networks, this model is designed to address the semantic bias issue caused by data augmentation in contrastive learning. Extensive experimental results on multiple real-world datasets (Beauty, Sports, and Toys) demonstrate that ACLSRec consistently and significantly surpasses a wide range of state-of-the-art benchmarks, including both attention-based and contrastive learning methods. The outstanding performance across all evaluation metrics highlights the effectiveness of our adaptive contrastive learning module in improving recommendation performance. This work demonstrates a promising paradigm of conducting contrastive learning with semantic consistency for sequential recommendation.

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A Appendix

A.1 The Learning Steps of ACLSRec

Algorithm 1: The Learning Steps of ACLSRec

Input: Batch of user sequences S, hyperparameters $\lambda, \beta, \gamma, \tau$.
Output: Updated model parameters Θ.
Step 1: User Sequence Encoding
1. For each sequence $s_i \in S$, embed the input sequence $s_i = \{v_1, v_2, \ldots, v_t\}$ into dense vectors: $\mathbf{H}_{in} = \text{EmbeddingLayer}(s_i)$
2. Encode $\boldsymbol{H}_{in}$ using a shared multi-head Transformer: $\boldsymbol{H}_o = \text{Transformer}(\boldsymbol{H}_{in})$
Step 2: Generate Perturbed Representation
3. Apply perturbation network MLP_{θ_p}: $\boldsymbol{H}_a = \text{MLP}_{\theta_p}(\boldsymbol{H}_{in})$
4. Encode $\boldsymbol{H}_a$ using a shared multi-head Transformer: $\boldsymbol{H}_p = \text{Transformer}(\boldsymbol{H}_a)$
Step 3: Restoration and Contrastive View Construction
5. Restore from perturbed representation using restoration network MLP_{θ_r}: $\boldsymbol{H}_{pr} = \text{MLP}_{\theta_r}(\boldsymbol{H}_p)$
6. Generate contrastive view from perturbation network $\text{MLP}_{\theta_{p'}}$: $\boldsymbol{H}_{op} = \text{MLP}_{\theta_{p'}}(\boldsymbol{H}_o)$.
Step 4: Multi-task Loss Computation
7.Compute main recommendation loss: $\mathcal{L}_{main}$ (Equation (32)).
8.Compute contrastive loss: $\mathcal{L}_{cl}$ (Equation (19)).
9. Compute alignment loss: $\mathcal{L}_{align}$ (Equation (23)).
10. Compute regularization term: R (Equation (31)).
11. Compute total loss: $\mathcal{L} = \mathcal{L}_{main} + \lambda\mathcal{L}_{cl} + \beta\mathcal{L}_{align} + \gamma R$
Step 5: Optimization
12. Backpropagate $\mathcal{L}$ to update model parameters.
return model parameters Θ.

We provides additional implementation details for the learning process of ACLSRec and outline the complete learning steps in Algorithm 1. The algorithm further details the computation and optimization of the combined loss function in a batch-training manner.

A.2 MLP Depth Analysis

To investigate the impact of Multi-Layer Perceptron (MLP) depth on model performance, we evaluated architectures with 1 to 4 layers on the Beauty and Sports datasets. As illustrated in Figure 5, the 2-layer MLP consistently achieves the optimal performance across all evaluation metrics. When the layer count increases to 3 or 4, we observe a systematic performance degradation across both datasets. This decline can be attributed to reduced gradient propagation efficiency and progressive information loss in excessively deep networks. These findings establish the 2-layer MLP as the optimal choice, achieving good learning outcomes while effectively mitigating overfitting risks.

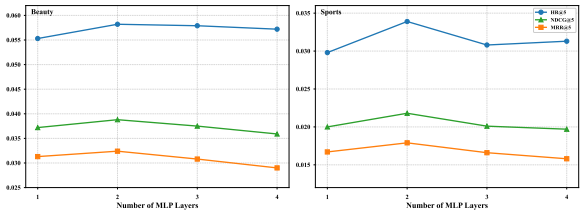


Figure 5: MLP Depth Analysis.